

EMCH 367 Controls

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Fall 2024

Homework #1

Due date: September 6, 11:59 pm

Problem 1. (20pt) True or False

1. Mathematical modeling of LTI system does not vary over time.
2. Principle of superposition holds for linear system.
3. Closed loop system does not need sensors.
4. Complementary solution of spring-damper system is given as an exponential function.
5. Mass-spring-damper system is a second order system.
6. Laplace transformation is a nonlinear operator.
7. Laplace transform of a unit step function is $1/s$.
8. Laplace transform of differentiation does not need initial conditions.
9. Final value theorem, $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$, is true for any $f(t)$.
10. Transfer function is the Laplace transform of the output to the Laplace transform of the input

Number	T/F	Number	T/F
1	T	6	F
2	T	7	T
3	F	8	F
4	T	9	F
5	T	10	T

Problem 2. (20pt) Transformation and Transfer Function

(a) (10pt) Find the Laplace transformation of

$$f(t) = \begin{cases} 0, & t < 0, \\ \sin(wt + \theta), & t \geq 0, \end{cases} \quad (1)$$

where $\theta \in \mathbb{R}$ is a constant.

Solution: First, note that

$$\sin(wt + \theta) = \sin wt \cdot \cos \theta + \cos wt \cdot \sin \theta,$$

and hence,

$$\begin{aligned} \mathcal{L}[f(t)] &= \mathcal{L}[\sin wt \cdot \cos \theta + \cos wt \cdot \sin \theta] \\ &= \cos \theta \mathcal{L}[\sin wt] + \sin \theta \mathcal{L}[\cos wt] \\ &= \cos \theta \frac{w}{s^2 + w^2} + \sin \theta \frac{s}{s^2 + w^2} \\ &= \frac{\sin \theta \cdot s + \cos \theta \cdot w}{s^2 + w^2}. \end{aligned}$$

(b) (10pt) Find the Laplace transformation of $f(t)$ given in Figure 1.

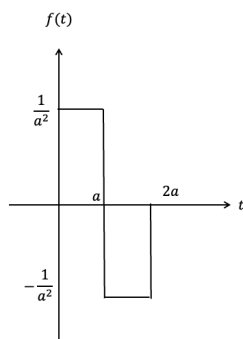


Figure 1

Solution: First, note that

$$f(t) = \frac{1}{a^2}1(t) - \frac{2}{a^2}1(t-a) + \frac{1}{a^2}1(t-2a),$$

and hence,

$$\begin{aligned}\mathcal{L}[f(t)] &= \frac{1}{a^2} \frac{1}{s} - \frac{2}{a^2} \frac{1}{s} e^{-as} + \frac{1}{a^2} \frac{1}{s} e^{-2as} \\ &= \frac{1 - 2e^{-as} + e^{-2as}}{a^2 s}.\end{aligned}$$

Problem 3. (20pt) Partial fraction expansion

(a) (10pt) Find the inverse Laplace transformation of the following function.

$$X(s) = \frac{1}{s(s+1)(s+2)(s+3)}$$

Solution: It follows from partial fraction expansion that

$$X(s) = \frac{a_1}{s} + \frac{a_2}{s+1} + \frac{a_3}{s+2} + \frac{a_4}{s+3}.$$

Now, it follows that

$$\begin{aligned} \frac{1}{(s+1)(s+2)(s+3)} &= a_1 + s \left[\frac{a_2}{s+1} + \frac{a_3}{s+2} + \frac{a_4}{s+3} \right] \implies a_1 = \frac{1}{6}, \\ \frac{1}{s(s+2)(s+3)} &= a_2 + (s+1) \left[\frac{a_1}{s} + \frac{a_3}{s+2} + \frac{a_4}{s+3} \right] \implies a_2 = -\frac{1}{2}, \\ \frac{1}{s(s+1)(s+3)} &= a_3 + (s+2) \left[\frac{a_1}{s} + \frac{a_2}{s+1} + \frac{a_4}{s+3} \right] \implies a_3 = \frac{1}{2}, \\ \frac{1}{s(s+1)(s+2)} &= a_4 + (s+3) \left[\frac{a_1}{s} + \frac{a_2}{s+1} + \frac{a_3}{s+2} \right] \implies a_4 = -\frac{1}{6}. \end{aligned}$$

Therefore, it follows that

$$X(s) = \frac{1/6}{s} - \frac{1/2}{s+1} + \frac{1/2}{s+2} - \frac{1/6}{s+3},$$

and hence,

$$x(t) = \frac{1}{6} - \frac{1}{2}e^{-t} + \frac{1}{2}e^{-2t} - \frac{1}{6}e^{-3t}.$$

(b) (10pt) Find the inverse Laplace transformation of the following function.

$$X(s) = \frac{1}{s^2(s+1)^2}$$

Solution: It follows from partial fraction expansion that

$$X(s) = \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s+1} + \frac{a_4}{(s+1)^2}.$$

Now, it follows that

$$\begin{aligned} \frac{1}{(s+1)^2} &= a_1 s + a_2 + s^2 \left[\frac{a_3}{s+1} + \frac{a_4}{(s+1)^2} \right] \implies a_2 = 1, \\ \frac{1}{s^2} &= a_3(s+1) + a_4 + (s+1)^2 \left[\frac{a_1}{s} + \frac{a_2}{s^2} \right] \implies a_4 = 1. \end{aligned}$$

Taking derivative will yield

$$\begin{aligned} -\frac{2}{(s+1)^3} &= a_1 + s[\dots] \implies a_1 = -2, \\ -\frac{2}{s^3} &= a_3 + (s+1)[\dots] \implies a_3 = 2. \end{aligned}$$

Therefore, it follows that

$$X(s) = \frac{-2}{s} + \frac{1}{s^2} + \frac{2}{s+1} + \frac{1}{(s+1)^2},$$

and hence,

$$x(t) = -2 + t + 2e^{-t} + te^{-t}.$$

Problem 4 (20pt) 1st order system

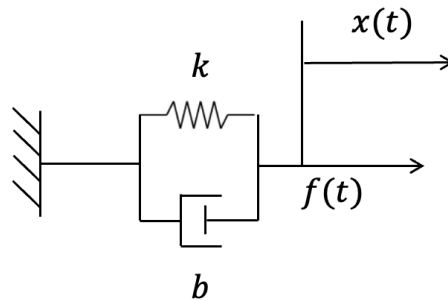
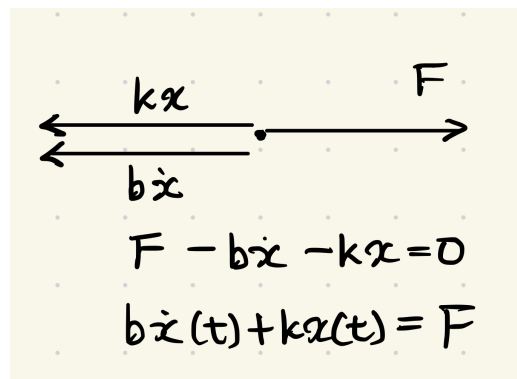


Figure 2

- (a) (5pt) Draw free body diagram and find the force balance equation in ODE.



- (b) (5pt) Derive the free vibration response.

Solution: The equation of motion is given by

$$b\dot{x}(t) + kx(t) = F(t), \quad x(0) = x_0,$$

For free vibration response, we have

$$b\dot{x}(t) + kx(t) = 0 \implies \frac{b}{k}\dot{x}(t) + x(t) = 0 \implies T\dot{x}(t) + x(t) = 0,$$

and the solution is given by

$$x(t) = x_0 e^{-\frac{1}{T}t}.$$

- (c) (8pt) Derive the forced vibration response with $f(t) = F(t)/k = \sin wt$ and $x(0) = 0$ using Laplace transformation. (hint: Define $T = b/k$, and consider T and w as constants.)

Solution: Note that the equation of motion is given as

$$T\dot{x}(t) + x(t) = \sin wt.$$

Taking Laplace transformation would yield

$$(Ts + 1)X(s) = \frac{w}{s^2 + w^2} \implies X(s) = \frac{w}{(s^2 + w^2)(Ts + 1)}.$$

Now, the partial fraction of the RHS is given as

$$\begin{aligned} \frac{a}{Ts + 1} + \frac{bs + c}{s^2 + w^2} &= \frac{(a + bT)s^2 + (b + cT)s + aw^2 + c}{(s^2 + w^2)(Ts + 1)} \\ &= \frac{w}{(s^2 + w^2)(Ts + 1)}. \end{aligned}$$

Therefore, we have

$$a + bT = 0, \quad b + cT = 0, \quad aw^2 + c = w.$$

Solving these three equations yields

$$a = \frac{T^2w}{1 + T^2w^2}, \quad b = -\frac{Tw}{1 + T^2w^2}, \quad c = \frac{w}{1 + T^2w^2}.$$

Therefore, it follows that

$$\begin{aligned} x(t) &= \mathcal{L}^{-1} \left[\frac{a}{T} \frac{1}{s + 1/T} + b \frac{s}{s^2 + w^2} + \frac{c}{w} \frac{w}{s^2 + w^2} \right] \\ &= \frac{Tw}{1 + T^2w^2} e^{-t/T} - \frac{Tw}{1 + T^2w^2} \cos wt + \frac{1}{1 + T^2w^2} \sin wt. \end{aligned}$$

- (d) (2pt) Let $k = 250$ N/m and $b = 400$ N·s/m. Calculate time constant.

Solution: Note that

$$T = \frac{b}{k} = \frac{8}{5} = 1.6 \text{ s}.$$

Problem 5. (20pt) 2nd order system

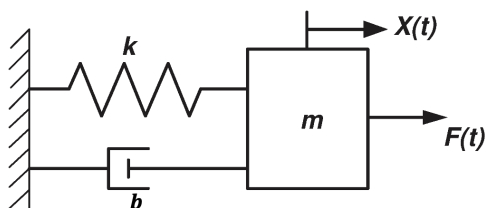


Figure 3

- (a) (5pt) Draw free body diagram and find the force balance equation in ODE.

$$m\ddot{x} = F - b\dot{x} - kx$$

$$m\ddot{x} + b\dot{x} + kx = F$$

- (b) (5pt) What is the inverse Laplace transformation of

$$\frac{s}{(s+a)^2}$$

Solution: Let $F(s) = \frac{1}{(s+a)^2}$ and note that

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{s}{(s+a)^2}\right] &= \mathcal{L}^{-1}[sF(s)] \\ &= \frac{d}{dt}f(t),\end{aligned}$$

and since $f(t) = te^{-at}$, it follows that

$$\begin{aligned}\mathcal{L}^{-1}\left[\frac{s}{(s+a)^2}\right] &= \frac{d}{dt}[te^{-at}] \\ &= e^{-at} - ate^{-at}.\end{aligned}$$

- (c) (5pt) Find the free vibration response with $b^2 - 4mk = 0$ by using Laplace transformation. The initial conditions are given by $x(0) = x_0$ and $\dot{x}(0) = \dot{x}_0$.

Solution: The equation of motion for free vibration is given by

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = 0, \quad x(0) = x_0, \quad \dot{x}(0) = \dot{x}_0.$$

The Laplace transformation will yield

$$(ms^2 + bs + k)X(s) - msx_0 - m\dot{x}_0 - bx_0 = 0,$$

and hence,

$$X(s) = \frac{x_0s + \dot{x}_0 + \frac{b}{m}x_0}{s^2 + \frac{b}{m}s + \frac{k}{m}}.$$

Since $b^2 - 4mk = 0$, it follows that $k/m = b^2/(4m^2)$. Therefore, we have

$$X(s) = \frac{x_0s}{(s + \frac{b}{2m})^2} + \frac{\dot{x}_0 + \frac{b}{m}x_0}{(s + \frac{b}{2m})^2},$$

and hence, taking inverse Laplace transformation will yield

$$x(t) = x_0 \left(e^{-\frac{b}{2m}t} - \frac{b}{2m}te^{-\frac{b}{2m}t} \right) + \left(\dot{x}_0 + \frac{b}{m}x_0 \right) te^{-\frac{b}{2m}t}.$$

- (d) (5pt) Let $m = 10$ kg, $k = 160$ N/m, $b = 64$ N·s/m, and $\omega = \pi/2$ s⁻¹. Calculate damped frequency of the system.

Solution: The natural frequency and damping ratio are given by

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{160}{10}} = 4 \text{ rad/s}$$
$$\zeta = \frac{b}{2\sqrt{mk}} = \frac{64}{2\sqrt{160 \cdot 10}} = \frac{64}{80} = 0.8.$$

Therefore, the damped frequency is given by

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 2.4 \text{ rad/s}$$

Bonus Problem. (20pt) Find the transfer function of the system given in Figure 4 wherein $x_i(t)$ is the input and $x_o(t)$ is the output. Assume that $x_i(0^-) = 0$ and $x_o(0^-) = 0$.

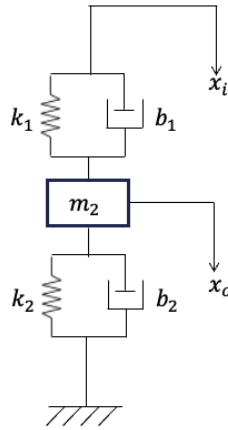


Figure 4

Solution: The equation of motion is given by

The free-body diagram shows the mass m_2 with four forces acting on it. Two forces point downwards: $k_1(x_i - x_o)$ and $b_1(\dot{x}_i - \dot{x}_o)$. Two forces point upwards: k_2x_o and $b_2\dot{x}_o$.

$$m_2\ddot{x}_o = b_1(\dot{x}_i - \dot{x}_o) + k_1(x_i - x_o) - b_2\dot{x}_o - k_2x_o.$$

The Laplace transformation is given by

$$[m_2s^2 + (b_1 + b_2)s + (k_1 + k_2)]X_o(s) = (b_1s + k_1)X_i(s)$$

and hence, the transfer function is given by

$$\frac{X_o(s)}{X_i(s)} = \frac{b_1s + k_1}{m_2s^2 + (b_1 + b_2)s + k_1 + k_2}.$$

Look up table

	$f(t)$	$F(s)$
1	$\delta(t)$	1
2	$1(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	$\frac{t^{n-1}}{(n-1)!}$	$\frac{1}{s^n}$
5	t^n	$\frac{n!}{s^{n+1}}$
6	e^{-at}	$\frac{1}{s+a}$
7	te^{-at}	$\frac{1}{(s+a)^2}$
8	$\frac{1}{(n-1)!}t^{n-1}e^{-at}$	$\frac{1}{(s+a)^n}$
9	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
10	$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
11	$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
12	$\sinh \omega t$	$\frac{\omega}{s^2-\omega^2}$
13	$\cosh \omega t$	$\frac{s}{s^2-\omega^2}$
14	$\frac{1}{a}(1 - e^{-at})$	$\frac{1}{s(s+a)}$
15	$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
16	$\frac{1}{b-a}(be^{-bt} - ae^{-at})$	$\frac{s}{(s+a)(s+b)}$
17	$\frac{1}{ab} \left[1 + \frac{1}{a-b}(be^{-at} - ae^{-bt}) \right]$	$\frac{1}{s(s+a)(s+b)}$
18	$\frac{1}{a^2}(1 - e^{-at} - ate^{-at})$	$\frac{1}{s(s+a)^2}$
19	$\frac{1}{a^2}(at - 1 + e^{-at})$	$\frac{1}{s^2(s+a)}$
20	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$

	$f(t)$	$F(s)$
21	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
22	$\frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2} t$	$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
23	$-\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t - \varphi)$ $\varphi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{s}{s^2+2\zeta\omega_n s+\omega_n^2}$
24	$1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t + \varphi)$ $\varphi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$
25	$1 - \cos \omega t$	$\frac{\omega^2}{s(s^2+\omega^2)}$
26	$\omega t - \sin \omega t$	$\frac{\omega^2}{s^2(s^2+\omega^2)}$
27	$\sin \omega t - \omega t \cos \omega t$	$\frac{2\omega^2}{(s^2+\omega^2)^2}$
28	$\frac{1}{2\omega} t \sin \omega t$	$\frac{s}{(s^2+\omega^2)^2}$
29	$t \cos \omega t$	$\frac{s^2-\omega^2}{(s^2+\omega^2)^2}$
30	$\frac{1}{\omega_2^2-\omega_1^2} (\cos \omega_1 t - \cos \omega_2 t), \quad \omega_1 \neq \omega_2$	$\frac{s}{(s^2+\omega_1^2)(s^2+\omega_2^2)}$
31	$\frac{1}{2\omega} (\sin \omega t + \omega t \cos \omega t)$	$\frac{s^2}{(s^2+\omega^2)^2}$